

$x[n]$

n samples

Feasibility of Image Processing in Frequency Domain

- Filtering an $M \times N$ image by a $m \times n$ kernel

Filtering	Order of operations
Spatial filtering	$MNmn$
Spatial filtering with separable kernel	$MN(m+n)$
Filtering in frequency domain using FFT	$2MN \log_2 MN$

- Computational advantage (assuming M & m)

- over spatial filtering:

$$C_n(m) = \frac{M^2 m^2}{2M^2 \log_2 M^2} = \frac{m^2}{4 \log_2 M} = \frac{(10)^2}{4 \log_2 100}$$

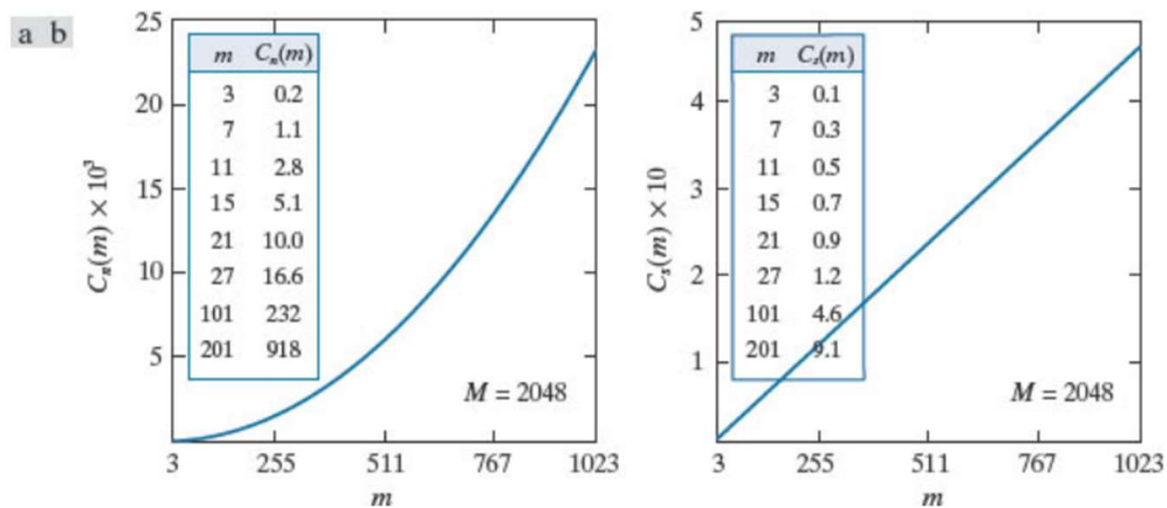
$m=10, M=100$

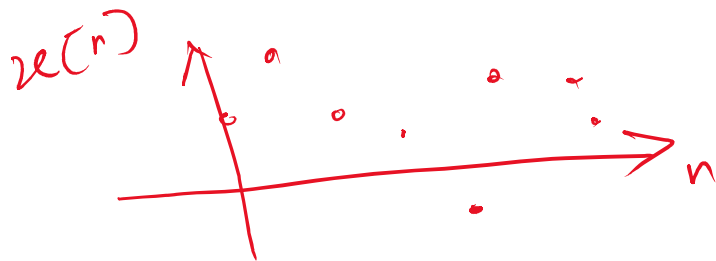
- over separable spatial filtering:

$$C_s(m) = \frac{2M^2 m}{2M^2 \log_2 M^2} = \frac{m}{2 \log_2 M}$$

Figure 4.2

(a) Computational advantage of the FFT over nonseparable spatial kernels. (b) Advantage over separable kernels. The numbers for $C(m)$ in the inset tables are not to be multiplied by the factors of 10 shown for the curves.





x has M samples $\rightarrow$ DFT requires M^2

FFT reduces that to $M \log_2 M$

- One variable (M samples)

$$F(u) = \sum_{x=0}^{M-1} f(x) e^{-j2\pi ux/M}$$

$u = 0, 1, 2, \dots, M-1$

$$f(x) = \frac{1}{M} \sum_{u=0}^{M-1} F(u) e^{j2\pi ux/M}$$

$x = 0, 1, 2, \dots, M-1$

- Two variables ($M \times N$ samples)

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$$

$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)}$$

$M = 100 \quad N = 10 \rightarrow$

$\frac{100}{66}$

DFT = 100

FFT = 66

RELATIONSHIP BETWEEN THE SAMPLING AND FREQUENCY INTERVALS

If $f(x)$ consists of M samples of a function $f(t)$ taken ΔT units apart, the length of the record comprising the set $\{f(x)\}$, $x = 0, 1, 2, \dots, M-1$, is

$$T = M\Delta T \quad (4-49)$$

The corresponding spacing, Δu , in the frequency domain follows from Eq. (4-41):

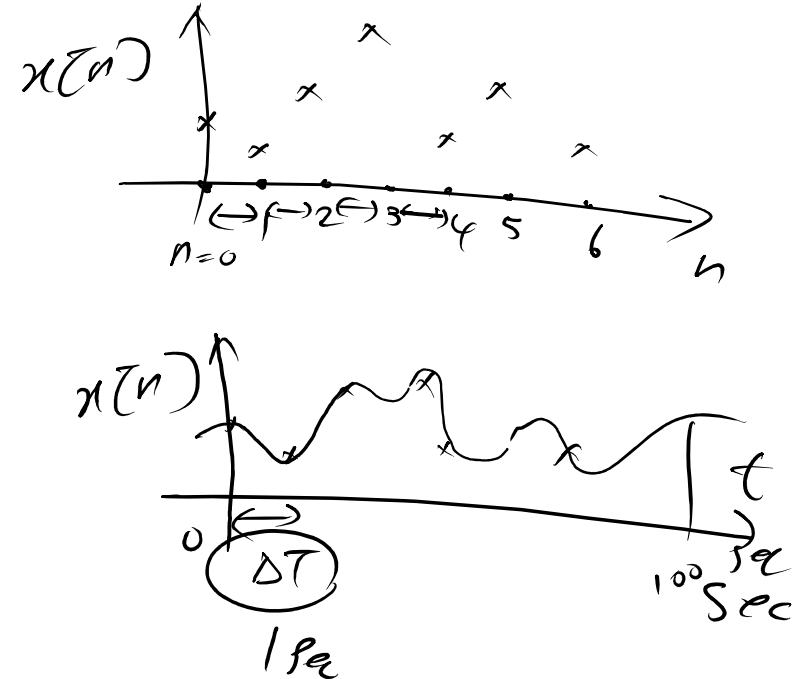
$$\Delta u = \frac{1}{M\Delta T} = \frac{1}{T} \quad (4-50)$$

4.4 The Discrete Fourier Transform of One Variable 229

The entire frequency range spanned by the M components of the DFT is then

$$R = M\Delta u = \frac{1}{\Delta T} \quad (4-51)$$

Thus, we see from Eqs. (4-50) and (4-51) that the resolution in frequency, Δu , of the DFT depends inversely on the length (duration, if t is time) of the record, T , over which the continuous function, $f(t)$, is sampled; and the range of frequencies spanned by the DFT depends on the sampling interval ΔT . Keep in mind these *inverse* relationships between Δu and ΔT .



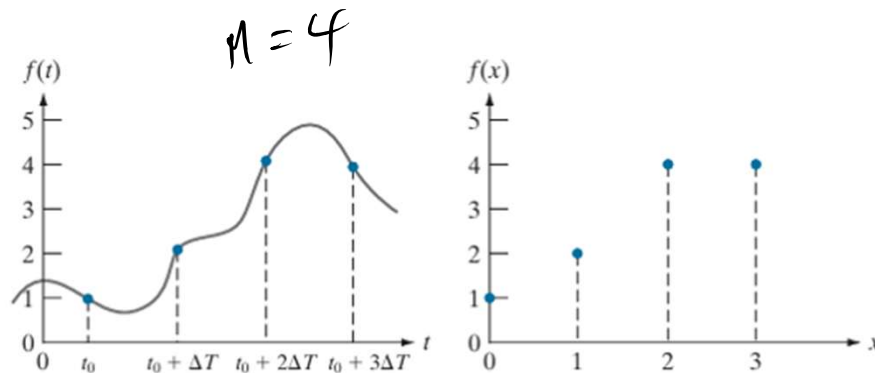
EXAMPLE 4.4: The mechanics of computing the DFT.

Figure 4.12(a) shows four samples of a continuous function, $f(t)$, taken ΔT units apart. Figure 4.12(b) shows the samples in the x -domain. The values of x are 0, 1, 2, and 3, which refer to the number of the samples in sequence, counting up from 0. For example, $f(2) = f(t_0 + 2\Delta T)$, the third sample of $f(t)$.

a b

FIGURE 4.12

(a) A continuous function sampled ΔT units apart.
(b) Samples in the x -domain.
Variable t is continuous, while x is discrete.



$$F(0) = \sum_{n=0}^3 f(n) = 1 + 2 + 4 + 4 = 11$$

$$F(1) = \sum_{n=0}^3 f(n) e^{-j 2\pi (1)n/4} = \underbrace{f(0)}_1 e^0 + \underbrace{f(1)}_2 e^{-j\pi/2} + \underbrace{f(2)}_4 e^{-j\pi} + \underbrace{f(3)}_4 e^{-j3\pi/2}$$

$$F(2) = -(1 + 0j), F(3) = -(3 + 2j)$$

$$= -3 + 2j$$

get back my $f(n)$ from $F(u)$

$$f(0) = \frac{1}{4} \sum_{u=0}^{M-1} F(u) e^{j2\pi(0)u} = \frac{1}{4} \sum_{u=0}^3 F(u)$$

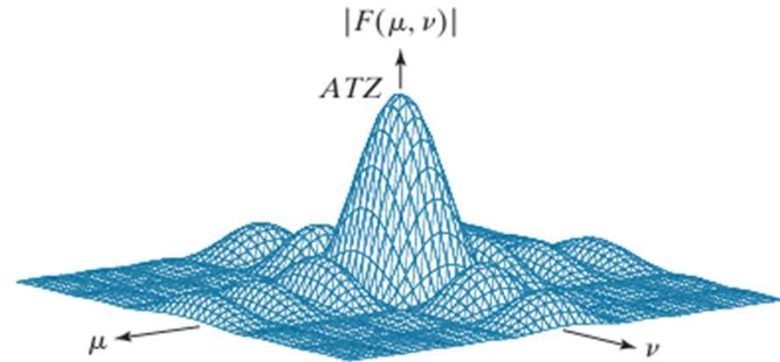
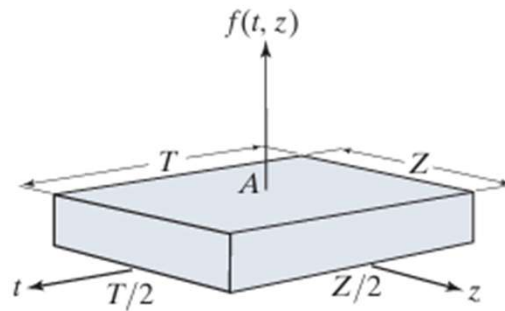
$$= \frac{1}{4} [11 - 3 + 2j - 1 - 3 - 2j] = \frac{1}{4} \times 4 = 1$$

0	1
2	4

a b

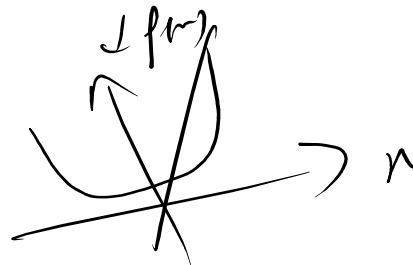
FIGURE 4.14

(a) A 2-D function and (b) a section of its spectrum. The box is longer along the t -axis, so the spectrum is more contracted along the μ -axis.



$$s_{\Delta T \Delta Z}(t, z) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \delta(t - m\Delta T, z - n\Delta Z) \quad (4-61)$$

$$y = f(n) = x = n^2$$



Periodicity & Centering

- The 2-D Fourier transform and its inverse are **infinitely periodic** in the u and v directions (as in 1-D):

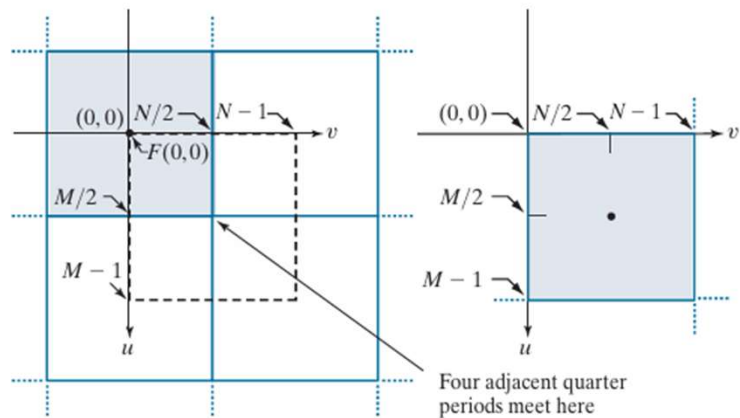
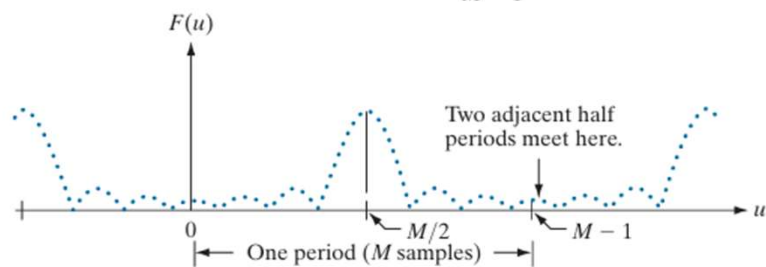
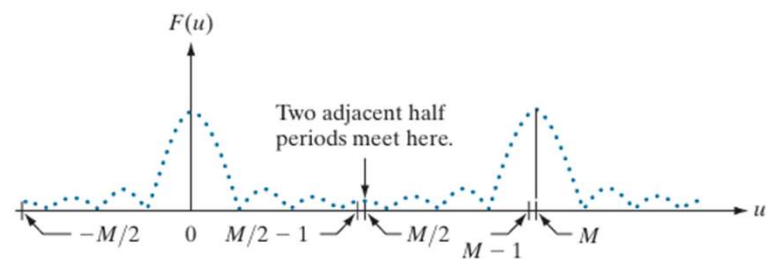
$$F(u, v) = F(u + k_1 M, v) = F(u, v + k_2 N) = F(u + k_1 M, v + k_2 N)$$

$$f(x, y) = f(x + k_1 M, y) = f(x, y + k_2 N) = f(x + k_1 M, y + k_2 N)$$

- Centering:

- Since $f(x)e^{j2\pi(u_0 x/M)} \Leftrightarrow F(u - u_0)$, we have $f(x)(-1)^x \Leftrightarrow F(u - M/2)$
- Multiplying $f(x)$ by $(-1)^x$ shifts the data, centering $F(u)$ on the interval $[0, M-1]$

The periodicities of the transform and its inverse are important issues in the implementation of DFT-based algorithms. Consider the 1-D spectrum in Fig. 4.22(a). As explained in Section 4.4 [see the footnote to Eq. (4-42)], the transform data in the interval from 0 to $M-1$ consists of two half periods meeting at point $M/2$, but with the lower part of the period appearing at higher frequencies. For display and filtering purposes, it is more convenient to have in this interval a complete period of the transform in which the data are contiguous and ordered properly,



-  = $M \times N$ data array computed by the DFT with $f(x, y)$ as input
-  = $M \times N$ data array computed by the DFT with $f(x, y)(-1)^{x+y}$ as input
-  = Periods of the DFT

Fourier Spectrum and Phase Angle

- In Polar Form: $F(u, v) = R(u, v) + jI(u, v) = |F(u, v)| e^{j\phi(u, v)}$
- Magnitude: $|F(u, v)| = [R^2(u, v) + I^2(u, v)]^{1/2}$
- Phase angle/spectrum: $\phi(u, v) = \arctan \left[\frac{I(u, v)}{R(u, v)} \right]$
- Power Spectrum: $P(u, v) = |F(u, v)|^2$
- DC Component: $F(0, 0) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) = MN \bar{f}$

Table 4.3

Summary of DFT definitions and corresponding expressions.

Name	Expression(s)
1) Discrete Fourier transform (DFT) of $f(x, y)$	$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$
2) Inverse discrete Fourier transform (IDFT) of $F(u, v)$	$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)}$
3) Spectrum	$ F(u, v) = [R^2(u, v) + I^2(u, v)]^{1/2} \quad R = \text{Real}(F); I = \text{Imag}(F)$
4) Phase angle	$\phi(u, v) = \tan^{-1} \left[\frac{I(u, v)}{R(u, v)} \right]$
5) Polar representation	$F(u, v) = F(u, v) e^{j\phi(u, v)}$
6) Power spectrum	$P(u, v) = F(u, v) ^2$
7) Average value	$\bar{f} = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) = \frac{1}{MN} F(0, 0)$
8) Periodicity (k_1 and k_2 are integers)	$F(u, v) = F(u + k_1 M, v) = F(u, v + k_2 N)$ $= F(u + k_1, v + k_2 N)$ $f(x, y) = f(x + k_1 M, y) = f(x, y + k_2 N)$ $= f(x + k_1 M, y + k_2 N)$
9) Convolution	$(f \star h)(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n) h(x - m, y - n)$
10) Correlation	$(f \star h)(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f^*(m, n) h(x + m, y + n)$
11) Separability	The 2-D DFT can be computed by computing 1-D DFT transforms along the rows (columns) of the image, followed by 1-D transforms along the columns (rows) of the result. See Section 4.11.
12) Obtaining the IDFT using a DFT algorithm	$MN f^*(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F^*(u, v) e^{-j2\pi(ux/M + vy/N)}$ This equation indicates that inputting $F^*(u, v)$ into an algorithm that computes the forward transform (right side of above equation) yields $MN f^*(x, y)$. Taking the complex conjugate and dividing by MN gives the desired inverse. See Section 4.11.

Table 4.4

Summary of DFT pairs. The closed-form expressions in 12 and 13 are valid only for continuous variables. They can be used with discrete variables by sampling the continuous expressions.

Name	DFT Pairs
1) Symmetry properties	See Table 4.1
2) Linearity	$af_1(x,y) + bf_2(x,y) \Leftrightarrow aF_1(u,v) + bF_2(u,v)$
3) Translation (general)	$f(x,y)e^{j2\pi(u_0x/M + v_0y/N)} \Leftrightarrow F(u - u_0, v - v_0)$ $f(x - x_0, y - y_0) \Leftrightarrow F(u,v)e^{-j2\pi(u x_0/M + v y_0/N)}$
4) Translation to center of the frequency rectangle, $(M/2, N/2)$	$f(x,y)(-1)^{x+y} \Leftrightarrow F(u - M/2, v - N/2)$ $f(x - M/2, y - N/2) \Leftrightarrow F(u,v)(-1)^{u+v}$
5) Rotation	$f(r, \theta + \theta_0) \Leftrightarrow F(\omega, \varphi + \theta_0)$ $r = \sqrt{x^2 + y^2} \quad \theta = \tan^{-1}(y/x) \quad \omega = \sqrt{u^2 + v^2} \quad \varphi = \tan^{-1}(v/u)$
6) Convolution theorem [†]	$f \star h(x,y) \Leftrightarrow (F \bullet H)(u,v)$ $(f \bullet h)(x,y) \Leftrightarrow (1/MN)[(F \star H)(u,v)]$
7) Correlation theorem [†]	$(f \star h)(x,y) \Leftrightarrow (F^* \bullet H)(u,v)$ $(f^* \bullet h)(x,y) \Leftrightarrow (1/MN)[(F \star H)(u,v)]$
8) Discrete unit impulse	$\delta(x,y) \Leftrightarrow 1$ $1 \Leftrightarrow MN\delta(u,v)$
9) Rectangle	$\text{rec}[a,b] \Leftrightarrow ab \frac{\sin(\pi ua)}{(\pi ua)} \frac{\sin(\pi vb)}{(\pi vb)} e^{-j\pi(ua+vb)}$
10) Sine	$\sin(2\pi u_0x/M + 2\pi v_0y/N) \Leftrightarrow \frac{jMN}{2} [\delta(u + u_0, v + v_0) - \delta(u - u_0, v - v_0)]$
11) Cosine	$\cos(2\pi u_0x/M + 2\pi v_0y/N) \Leftrightarrow \frac{1}{2} [\delta(u + u_0, v + v_0) + \delta(u - u_0, v - v_0)]$
The following Fourier transform pairs are derivable only for continuous variables, denoted as before by t and z for spatial variables and by μ and ν for frequency variables. These results can be used for DFT work by sampling the continuous forms.	
12) Differentiation (the expressions on the right assume that $f(\pm\infty, \pm\infty) = 0$.)	$\left(\frac{\partial}{\partial t}\right)^m \left(\frac{\partial}{\partial z}\right)^n f(t,z) \Leftrightarrow (j2\pi\mu)^m (j2\pi\nu)^n F(\mu,\nu)$ $\frac{\partial^m f(t,z)}{\partial t^m} \Leftrightarrow (j2\pi\mu)^m F(\mu,\nu); \quad \frac{\partial^n f(t,z)}{\partial z^n} \Leftrightarrow (j2\pi\nu)^n F(\mu,\nu)$
13) Gaussian	$A2\pi\sigma^2 e^{-2\pi^2\sigma^2(t^2+z^2)} \Leftrightarrow Ae^{-(\mu^2+\nu^2)/2\sigma^2} \quad (A \text{ is a constant})$